

T4SKFORCE ENTERPRISE

Quantum-Proof Security Scanner for Banking Systems

Architecture Breakdown

T4SKFORCE Enterprise is built as a classic three-tier system so that scanning logic, risk-scoring intelligence, and stored evidence stay cleanly separated and independently scalable. The Presentation Tier is a React/Next.js dashboard that analysts use to trigger scans and read results. The Application Tier is a Python/FastAPI service layer that performs the actual cryptographic scanning, quantum-risk scoring, and AI-based migration prediction. The Data Tier persists every scan's Cryptographic Bill of Materials (CBOM), score history, raw telemetry, and trained model artifacts so that results are auditable and trends can be tracked over time. The diagram and breakdown below walk through each tier and how they connect.

Architecture Diagram

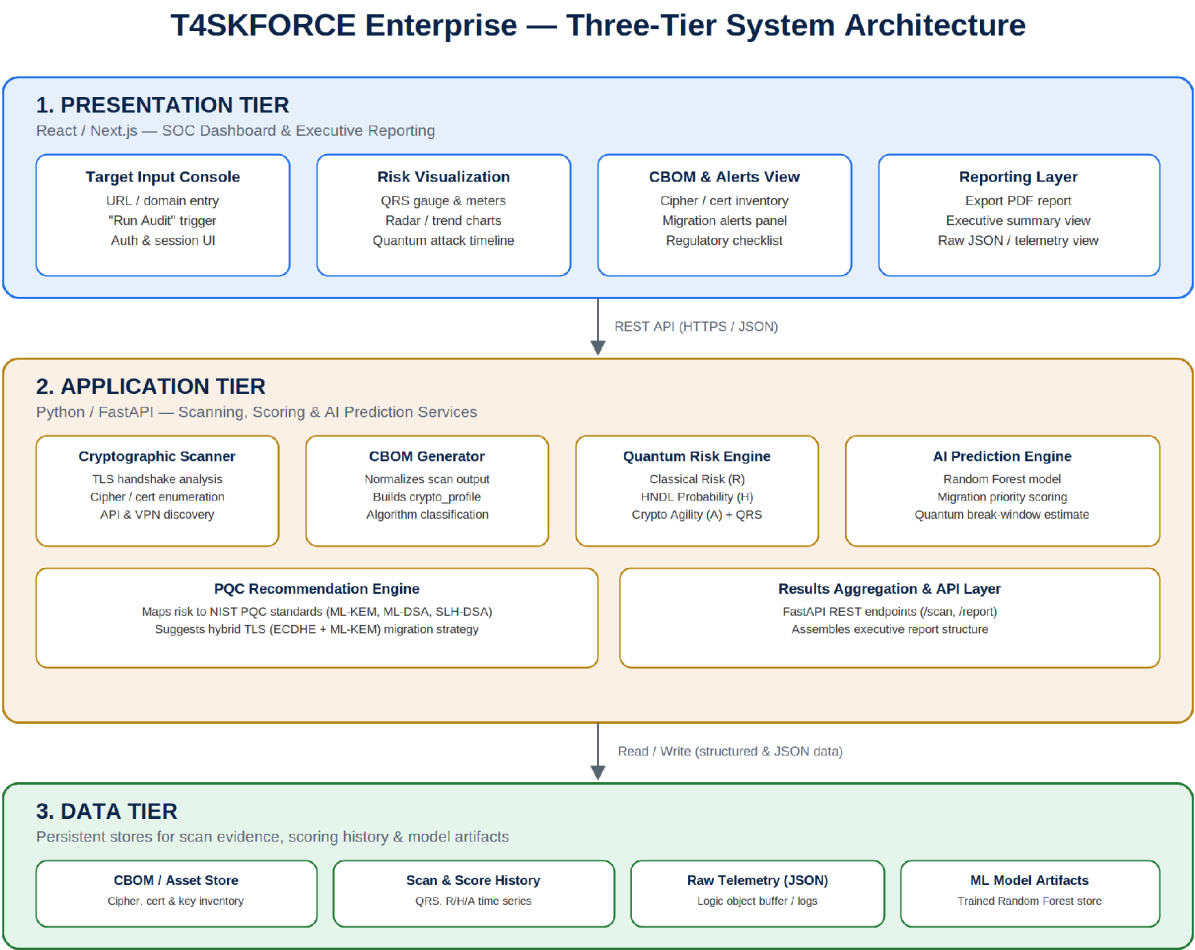


Figure 1. Three-tier architecture of T4SKFORCE Enterprise — Presentation, Application, and Data tiers.

1. Presentation Tier

The Presentation Tier is the analyst-facing SOC dashboard, built with React and Next.js. It is responsible for collecting the scan target from the user, calling the Application Tier's REST API, and rendering every result — scores, cryptographic inventory, and recommendations — in a readable, real-time interface. It holds no scoring logic itself; it only requests data from the Application Tier and displays it.

1.1 Components

Component	Responsibility
Target Input Console	Accepts a URL / domain / banking endpoint and triggers a new scan (“Run Audit”).
Risk Visualization	Renders the Quantum Readiness Score gauge, radar chart (risk vs. industry benchmark), and the quantum attack timeline.
CBOM & Alerts View	Displays the cipher suite / certificate / key-exchange inventory and flags migration alerts and regulatory gaps.
Reporting Layer	Exports the executive PDF report and exposes the raw JSON / telemetry view for technical users.

1.2 Technology & Design Notes

- Framework: React + Next.js for server-side rendering and fast dashboard load times.
- Communicates with the Application Tier exclusively over REST APIs (HTTPS / JSON) — no direct database access, which keeps the UI stateless and swappable.
- Visual elements: risk-score gauge, quantum-readiness meter, algorithm vulnerability chart, migration recommendations panel, and downloadable quantum security certificate.
- Designed for two audiences from the same views: SOC analysts (technical detail, raw telemetry) and executives (summarized risk posture, compliance status).

2. Application Tier

The Application Tier is the system's logic core, built in Python with FastAPI. It performs the cryptographic scan, converts raw findings into a structured CBOM, runs the Quantum Risk Engine to compute the Quantum Readiness Score, applies an AI model to predict migration urgency, and produces PQC migration recommendations. This tier is stateless per request — it reads from and writes to the Data Tier rather than holding data itself.

2.1 Components

Component	Responsibility
Cryptographic Scanner	Performs the TLS handshake, enumerates cipher suites, extracts certificate metadata, and discovers exposed APIs / VPN endpoints.
CBOM Generator	Normalizes raw scan output into a structured “crypto_profile” and classifies algorithms as Quantum Vulnerable, Quantum Resistant, or Deprecated.
Quantum Risk Engine	Computes Classical Risk (R), HNDL Probability (H), Crypto Agility (A), penalties, and the final Quantum Readiness Score.
AI Prediction Engine	A Random Forest model that outputs migration priority and an estimated quantum break-window from the extracted cryptographic features.

Component	Responsibility
PQC Recommendation Engine	Maps detected weaknesses to NIST PQC standards (ML-KEM, ML-DSA, SLH-DSA) and proposes a hybrid TLS migration strategy.
Results Aggregation / API Layer	FastAPI REST endpoints (e.g. /scan, /report) that assemble the final report structure and serve it to the Presentation Tier.

2.2 Technology & Design Notes

- Framework: FastAPI for a high-performance, asynchronous backend API.
- Risk scoring is a transparent, rule-based weighted model (not a black box), so every score is explainable back to the underlying CBOM evidence.
- AI component (scikit-learn Random Forest) is used only for predictive elements — migration priority and break-window estimation — layered on top of the deterministic risk score.
- Modular internal structure (Scanner → CBOM → Risk Engine → AI Engine → Recommendation Engine) allows each module to be updated independently as PQC standards evolve.

3. Data Tier

The Data Tier persists everything the Application Tier produces so that results survive beyond a single session, can be audited, and can be trended over time. It stores structured cryptographic inventories, historical scores, raw scan telemetry, and the trained ML model artifacts used by the AI Prediction Engine.

3.1 Components

Component	Responsibility
CBOM / Asset Store	Structured inventory of TLS versions, cipher suites, key-exchange algorithms, certificates, and key sizes discovered per scanned asset.
Scan & Score History	Time-series record of Quantum Readiness Score and its sub-scores (R, H, A) per asset, enabling trend and drift tracking.
Raw Telemetry (JSON)	The unprocessed “logic object buffer” — scan metadata, asset details, and protocol data — kept for audit and re-scoring.
ML Model Artifacts	Serialized, versioned Random Forest models used by the AI Prediction Engine, so predictions stay reproducible.

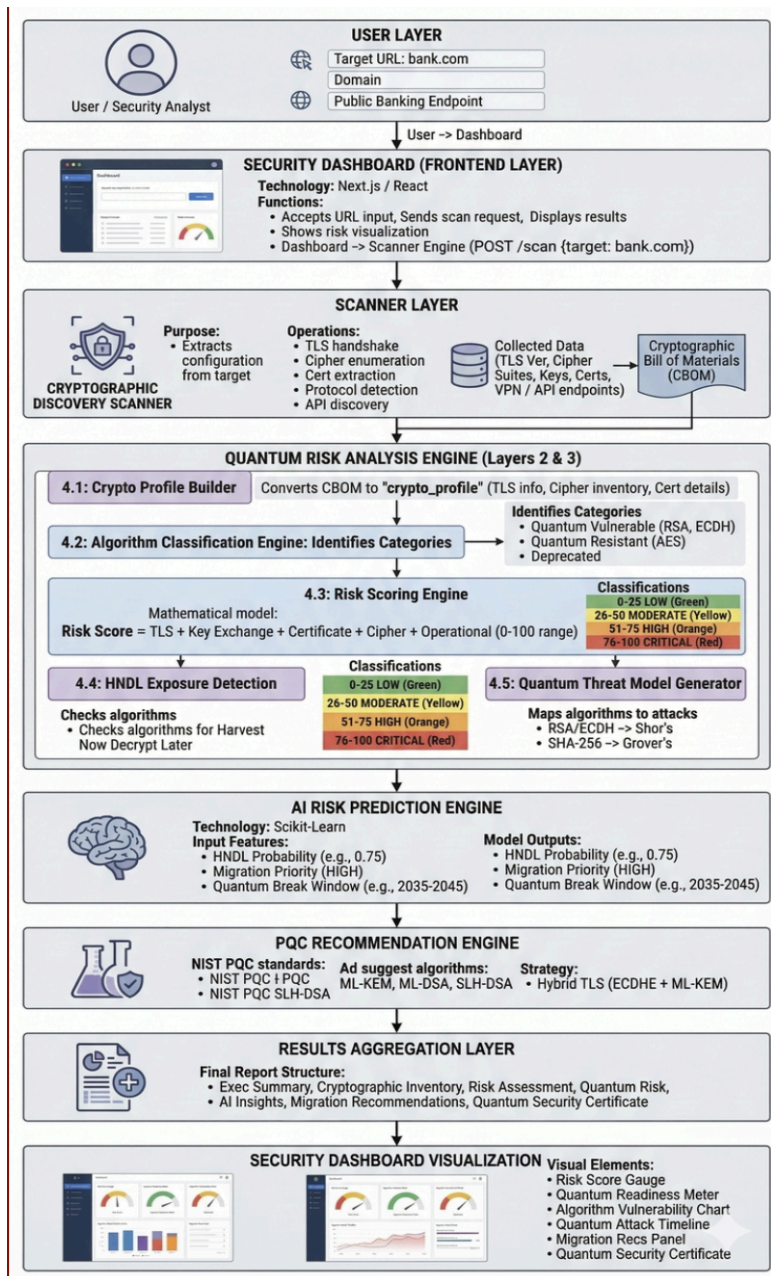
3.2 Technology & Design Notes

- Data is written only by the Application Tier; the Presentation Tier never touches the Data Tier directly, preserving a clean separation of concerns.
- Structured storage (CBOM, score history) and semi-structured JSON telemetry are kept separately so structured data can be queried for dashboards while raw telemetry remains available for deep audits.
- Historical score storage is what powers trend charts and the quantum attack timeline shown in the Presentation Tier.

4. How the Tiers Connect — End-to-End Flow

1. Analyst submits a target URL through the Presentation Tier.

- 2. Presentation Tier calls the Application Tier's REST API with the scan request.
- 3. Application Tier's Scanner performs the TLS/cipher/certificate scan and hands raw findings to the CBOM Generator.
- 4. CBOM Generator writes the structured cryptographic profile to the Data Tier and passes it to the Quantum Risk Engine.
- 5. Quantum Risk Engine computes R, H, A and the final Quantum Readiness Score; the AI Prediction Engine adds migration priority and break-window estimates.
- 6. PQC Recommendation Engine attaches migration guidance; the Results Aggregation layer assembles the full report and stores score history back to the Data Tier.
- 7. Presentation Tier fetches the aggregated result and renders the dashboard, alerts, and exportable PDF report.



5. Technology Stack by Tier

Tier	Core Technology	Primary Role
Presentation	React / Next.js	SOC dashboard, visualizations, PDF export
Application	Python, FastAPI, scikit-learn (Random Forest)	Scanning, CBOM generation, quantum risk scoring, AI prediction, PQC recommendations
Data	Structured CBOM store, score-history store, JSON telemetry store, model artifact store	Persistence, audit trail, trend history, reproducible AI predictions

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Demo video link:  **T4SKFORCE_Demo.mp4**